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Contextual Areas

Pricing and Assortment Strategies with Product Exchanges

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Abstract. Lenient return policies enable consumers to return or exchange products they are unsatisfied with, which boosts sales. Unfortunately, they also increase retailer costs. We develop a search framework where consumers sequentially learn about products' true value and evaluate whether to keep, exchange, or return them. Our formulation results in a tractable attraction demand model that can be used for optimization. We show that when pricing is not a decision, the assortment problem does not have a simple structure, but we provide an approximation algorithm to solve it. When prices and assortment can be controlled, the optimization becomes tractable: product prices can either be set so that potential return costs are added to the product price, be reduced to ensure that consumers choose to evaluate them after an exchange, or be set so high so that the items are effectively excluded from the assortment. We find that when prices and assortment can be jointly optimized, assortment size always increases when consumers pay a higher share of the return cost. Finally, retailers prefer to pass all return costs on to the consumers, which not only improves social welfare but also can raise consumer surplus.

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1. Introduction

In 2012, Zalando, a leading German online retail company, promoted itself with the slogan “Schrei vor Glück! Oder schicks zurück” (Scream for joy! Or send it back), which resulted in return rates of over 50% (Kontio 2012). Zalando later shortened it to simply “Schrei vor Glück!” (Scream for joy!), although it still offers very lenient return policies (Martell 2013). Zalando's case illustrates a general concern in the industry: the costs associated with handling returns amount to \$260.5 billion in 2015 in the United States alone, equal to 8.3% of total retail sales (Weinswig 2016). Despite these costs, most retailers acknowledge returns as a necessary evil. Indeed, offering the option of a return boosts sales, which may result in returns but may also lead to actual purchases. In particular, they help consumers accept (fit or functional) uncertainty about the product at the time of purchase. And even in the case of a return, consumers may not leave the retailer empty-handed but instead continue trying products to satisfy their consumption need.

For example, consider a (female) consumer who needs to buy a new mattress. Prior to purchase, the consumer can either go to a brick-and-mortar store and try several alternative mattress types (e.g., water, latex, futon), in different sizes (e.g., twin, queen, king)

and of different firmness levels (e.g., firm, medium, soft), for a few minutes, or inspect the product in an online shop. In either case, whether trying a mattress in person or inspecting it online, the consumer is left with a high level of uncertainty regarding the product's suitability. Mattress retailers have come to realize that this risk is important for the consumer, and they have responded by offering in-home trial periods. These allow consumers to resolve the uncertainty over a period of several weeks, with the option of returning or exchanging the product if they are not entirely satisfied. A consumer is thus able to repeatedly buy other products in the assortment until she finds the one that best meets her requirements.

The key elements of this example are that (1) product uncertainty is resolved after purchase and (2) return policies allow consumers to sequentially search through the assortment. This is an intuitive behavior when products are bulky or of high value. Of course, other consumer patterns are also possible, such as purchasing multiple items at once to just keep a few, depending on the product and industry characteristics, but the focus of this paper is on sequential search through returns and exchanges. Evidence of sequential consumer exchange patterns can be found in Samorani et al. (2018) for the electronics industry.

The authors not only found that over 38.7% of consumers continued shopping once they had returned the first product but also that some of them kept exchanging up to 4—and in rare cases, even 11—times. This indicates that upon return, consumers do not necessarily leave the retailer empty-handed. Instead, they sequentially search (by means of consecutive exchanges) through the assortment when uncertainty can be resolved only after purchase. Furthermore, the conditional probability of an exchange, after a return, was shown in Samorani et al. (2018) to increase as the search progressed (e.g., upon a second return, the probability of an exchange went from 38.7% to 56.1%). Thus returns and exchanges are just an element of an integral shopping experience for the consumer, which these authors call “episodes.” More important, their empirical analysis suggests that some of the common wisdom in the study of returns could be reversed when one considers episodic data instead of transactional data.

From this perspective, encouraging consumers to exchange a product is of great value to retailers, as it converts otherwise lost sales into potential revenue. On the other hand, product returns and exchanges are of considerable expense, in the form of shipping, sorting, and cleaning costs, among others. Retailers have taken various steps to compensate for this expense by claiming restocking fees or return shipping costs. For example, US-Mattress charges a restocking fee of up to \$139 (depending on the model), whereas start-up companies such as Casper offer free returns, and Mattress1One imposes a \$130 charge for transportation costs (Sleep Like the Dead 2018). Return policies vary across industries and are commonly charged by a wide variety of companies, as shown by Davis et al. (1998), whose survey documents variation in consumer return costs across retailers. To be able to suggest the appropriate return policy for a retailer, we need to have a comprehensive understanding of how such policy affects consumer behavior and, in particular, the consumer’s discovery process of the assortment. This is precisely the objective of this paper.

1.1. Contributions

In this research, we provide a novel framework of consumer search, where return and exchange decisions depend on a retailer’s assortment, pricing, and return fee conditions. We explicitly incorporate the fact that consumers face uncertainty about the true value of products before purchase and can only learn it by acquiring them. Specifically, they first form a preference list. Then they purchase a product and, after learning its true value, return or keep it; when it is returned, the consumer can walk away or choose the next product on her preference list for purchase, which means an exchange. The process repeats itself

until a product is kept or the consumer leaves after a return without any further purchase.

We solve the consumer problem using dynamic programming (DP). Assuming that product uncertainties are logistic distributed, we are able to write the resulting choice probabilities in an appealingly compact form, following an attraction model where product attractiveness depends on expected product valuation, rank in the preference list, and return costs charged to the consumer. Interestingly, consumers may not explore all available options; that is, the sequential exploration leads to a consideration set that may be strictly smaller than the entire assortment.

After specifying the demand model, we are ready to analyze the (male) retailer’s problem regarding the optimal assortment and pricing, as well as the most appropriate return policy. We approach these questions using two scenarios in the tractable case where return costs are product independent.¹ First, we assume that prices are exogenously fixed. In such a setting, we find that the assortment planning problem does not have a simple structure when returns are costly. Specifically, it can be optimal to skip some products that generate more revenue (Talluri and Van Ryzin 2004) or are preferred by consumers (Anderson et al. 1992) and in their place include products with a smaller margin and lower popularity. We nevertheless provide an algorithm based on dynamic programming that finds a solution within a factor of ϵ from the optimal solution and runs fast in practice. Second, we solve the joint pricing and assortment problem. In such a setting, the retailer always includes the most popular products (those with higher product value for the consumer) with a nontrivial pricing scheme: it is optimal (i) to hide the retailer’s share of the return cost in the price if possible, (ii) to set prices so that the consumer still includes the product in the consideration set although she is indifferent between buying it or waking away, or (iii) to set prices high enough so that the product is effectively excluded from consideration. We derive closed-form formulas to determine such optimal prices. Furthermore, it is optimal for the retailer to charge the consumer the entire return costs as not doing so limits the assortment size. It also maximizes social welfare and may even increase consumer surplus when more variety is offered.

The rest of the paper is organized as follows. Section 2 reviews the relevant literature. We then present our model in Section 3 and results in Section 4. Section 5 concludes. All proofs are relegated to the e-companion.

2. Literature Review

Our research is closely related to the two existing operations and marketing streams of literature: (1) empirical and analytical models of returns and

exchanges and (2) assortment planning and pricing models with consumer search.

As noted in the introduction, product returns are widespread and, as a result, have received attention from both empirical and analytical researchers. First, the empirical literature has provided insights on the drivers and the consequences of product returns. In trying to understand the reasons why consumers may be more likely to return a product, it has been found that retailer policies regarding fees are important (Wood 2001). In addition, more expensive items tend to be returned more (Hess and Mayhew 1997, Anderson et al. 2009). Returns also seem to increase when there is less information about the product (De et al. 2013), when there is more information about alternative products (Bechwati and Siegal 2005), when the assortment size is larger (Samorani et al. 2018), or in context-specific situations such as holiday seasons or when a gift card is used (Petersen and Kumar 2009). These studies suggest that return rates can be influenced by retailer decisions, so they can be embedded in optimization models, as we do here. In comparison, the consequences of product return on future purchase intentions have received less attention. Indeed, returns are just one more interaction between consumer and retailer in the context of a long consumer–retailer relationship. Bower and Maxham (2012) discuss the effect of return policies on future purchase intentions, whereas Petersen and Kumar (2009) and Griffis et al. (2012) evaluate whether consumers who have returned a product may be more likely to purchase in the future. These papers, however, treat future purchases as seemingly unrelated observations and therefore do not mention whether a future purchase is motivated by the desire to fulfill the same need (which is still unmet because the product has been returned) or whether it is triggered by a new one. This distinction is explicitly considered in Samorani et al. (2018), who conceptualize product return as an unsatisfying choice within the consumer's search through an assortment. In this framework, the authors confirm that higher-priced items lead to higher returns and that even though a larger assortment size increases the likelihood of returning an item, it also increases the likelihood of a consumer finally keeping it. We take a similar view and specifically build a consumer search model that retains the empirical features of Samorani et al. (2018).

The theoretical research on product returns has devoted much attention to analyzing the optimal pricing and return policies for a single-product monopolist (Hess et al. 1996) or two-product monopolists (Shulman et al. 2009, 2010). In particular, Shulman et al. (2009) allow heterogeneous consumers to either (i) keep and (once) exchange, assuming that the second product will always satisfy the consumer (a similar assumption is made in Shulman et al. 2010); (ii) keep and return; or

(iii) not purchase a product. We depart from this modeling framework in several ways. First, we allow consumers to repeatedly exchange products if they are not satisfied, a modeling feature that extends previous analysis of product returns to a dynamic setting where consumers potentially repeatedly return and exchange products if they are not satisfied. Secondly, analyzing a multiproduct setting allows us to discover the effect of product returns on retailer's optimal assortment choice. We are aware of only one paper that analytically examines the optimal assortment when consumers are allowed to return products: Alptekinoğlu and Grasas (2014) investigate the impact of return policies on the optimal assortment selection when exchanges are not considered. The presence of product returns fundamentally changes the assortment problem (e.g., it may be optimal to choose an assortment that is not made of the most popular items when prices are identical). By contrast, we build a model with the same objective of characterizing optimal retailer policies but with one main difference with these papers: we consider a multiple product setting with returns and exchange possibilities, including, for instance, long sequences of consumer–retailer interactions (e.g., purchase-exchange-purchase-return). This requires us to develop a framework where tractable demand functions for each product can be obtained and, in particular, one that allows prices and return fees to be easily optimized, in contrast with Alptekinoğlu and Grasas (2014).

In this sense, our approach also takes inspiration from search models in the economics literature. Weitzman (1979), for instance, builds a sequential consumer search model to (i) analyze the optimal order in which products are to be explored and (ii) determine the best point at which to stop searching. In his model, as in ours, it is assumed that each product has not only an exploration cost but also a reward associated with it. Although Weitzman's model is arguably more general than ours in allowing consumers to fall back on their preferred reward (recall or recourse) and in considering arbitrary reward distributions, our goal is to determine the demand distribution for each product in a tractable way intended for use in optimization. We therefore depart from Weitzman's consumer search framework by assuming (i) a particular demand distribution² and (ii) no possibility of recall.³

Recently, researchers have been able to integrate consumers' search behavior into retailers' optimal decision making. Najafi et al. (2016), for example, optimize retailers' prices when product quality is uncertain. In an assortment setting, Bernstein and Martínez-de-Albéniz (2017) analyze the optimal product rotation strategy when consumers can strategically visit stores multiple times. The key for consumers is to decide

when it is best to stop searching and when they should continue exploring the remaining options. In the presence of search costs, it is typically optimal to contain the search to the items within the *consideration set*. For example, Liu and Dukes (2013) analyze a multiproduct, multifirm setting where consumers can decide which firm and how many products of each firm to consider. Other recent papers that also incorporate this feature are Wang and Sahin (2018), Feldman and Topaloglu (2017), and Aouad et al. (2015). Closest to our work is Wang and Sahin (2018): they use the multinomial logit (MNL) demand model where the consumer, prior to choosing a product, forms her consideration set. They find that neither the revenue-ordered assortment (Talluri and Van Ryzin 2004) nor the same-price policy (Anderson et al. 1992) will remain optimal but find that it is optimal to apply a quasi-same pricing strategy. By contrast, we develop a sequential search framework with returns that leads to an attraction model for demand in which return costs influence product attractiveness and also impact retailer profitability.

Finally, our work is also related to the joint assortment planning and pricing problem. Anderson et al. (1992) show that it is optimal to use the same markup (price minus cost) for all products (although this may no longer hold when there are different price-sensitivity coefficients; see Wang 2012 or Gallego and Wang 2014). Under equal margins, it is optimal to offer the maximum variety when product costs are fixed (Heese and Martínez-de Albéniz 2018). In our case, however, we find that, even when the optimal pricing is such that products generate equal margins, it may be optimal to reduce variety so that return costs for the retailer are decreased.

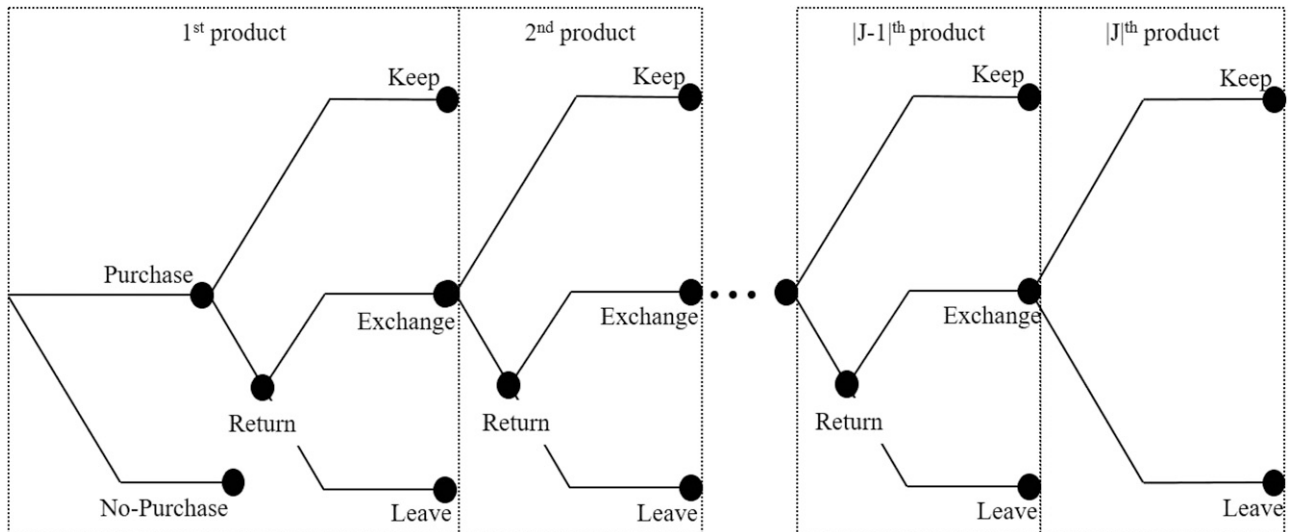
3. Model

We consider a one-time interaction between a retailer and a consumer. By *one-time interaction*, we mean that the consumer is interested in buying at most one product from a stable assortment shown by the retailer, and to do so, she may engage in a sequence of actions that will lead to either a purchase decision or the decision to leave the system without a transaction. This sequence will take the form of product choices (e.g., purchase product j) followed by keep, return, or exchange decisions, as shown in Figure 1. In this setting, the retailer will first decide on the assortment (which will remain fixed during the interaction with the consumer), the pricing of each item, and possible fees for every return. With this information, the consumer will choose her optimal purchase, exchange, and return strategy. We are ultimately interested in understanding how such consumer behavior affects the retailer's optimal assortment, pricing, and return fee decisions.

3.1. Model Setup

Consider a retailer who is planning his assortment for a certain product type (e.g., mattresses, electronics). He can choose to offer any of the products available in $N = \{1, 2, \dots, n\}$. We denote the retailer's assortment choice by $J \subseteq N$. Each product provides an expected value equal to u_j , $j \in N$, which differs across products but is identical across consumers. In other words, we consider consumers who are homogeneous in the expected valuation of products. Without loss of generality, we assume that unit production costs are set to zero (such costs can be incorporated in the value u_j ; see Heese and Martínez-de Albéniz 2018 for an example of how to do this), and the outside

Figure 1. Sequence of Choices Made by the Consumer



option (i.e., choosing not to purchase any product) provides a value for the consumer equal to $u_0 = 0$. Here, we implicitly assume that the outside option is deterministic and constant, which is reasonable when consumers have a good understanding of the value of leaving the retailer empty-handed, independent of whether a return was involved (e.g., the utility of not having a mattress at home).

Although this is a limitation of our model, one could also want to incorporate a random outside option where consumers experience an outside shock every time they decide to keep a product. Including such randomness is possible when outside options are highly correlated, and it results in a model with a mixture of customer types; on the other hand, when outside options are independent and identically distributed (i.i.d.), then the structure of customer choice probabilities remains similar to the one derived in our base model.

In addition to making assortment decisions, the retailer determines the price vector $\mathbf{r} = (r_1, \dots, r_{|J|})^4$ for all products made available in J . Thus, a consumer entering the retailer's store can choose to buy one item from the available assortment. The net expected utility derived from product $j \in J$ is given by $w_j := u_j - r_j$. Because product value may actually be different from the consumer's expectation w_j , we let the stochastic utility of product j be given as

$$U_j = w_j + \xi_j. \quad (1)$$

Here, ξ_j is a random shock observed after purchasing item j , one which alters the value of keeping product j , as opposed to returning it. For tractability, we assume that the uncertainty shock ξ_j is an i.i.d. logistic random variable with mean zero and scaling parameter 1; that is, its cumulative distribution function (c.d.f.) is $\Pr[\xi_j \leq x] = e^x / (1 + e^x)$. This assumption allows us to obtain closed-form demand functions.⁵

Such a random shock, prevalent in all the literature on returns, may occur because consumers are still uncertain about the product match after searching the retailer's store. For instance, because online consumers cannot touch or try a product in an e-commerce store prior to purchase, they may realize after delivery that it did not provide the expected functionality. Similarly, in brick-and-mortar stores, consumers may be exposed to disconfirming information and induce them to return a product (Bechwati and Siegal 2005). This will occur when U_j is lower than the expected utility derived from going back to the store to return or exchange the item, net of return fees. Indeed, returning a product is costly for both the retailer and the consumer. Return costs may be financial, including shipping fees, sorting, refurbishing, and restocking, or nonfinancial, including time spent

returning products. In addition, these costs can be product dependent (such as restocking fees proportional to product price) or fixed (hassle cost or shipping cost). To reflect this range of return costs, we denote the retailer's and consumer's costs as c_j and f_j , respectively. This general formulation is unfortunately not tractable when considering the retailer's problem, so later in Section 4 we must assume that return costs are product and price independent.

Because of the shocks to product utility, the consumer may engage in a sequence of visits to the retailer. In each one of these visits, she will compare the products' net expected utilities w_j and choose to either buy a product or return it. Figure 1 illustrates the possibilities that the consumer will face during this sequence.

Finally, we assume a model without recall. In other words, once a product is returned, the consumer cannot purchase it again, so it is excluded from the future assortment, as in Bernstein and Martínez-de-Albéniz (2017).⁶ As a result of this assumption, the product exchange sequence may continue until every item to the consumer's liking has been returned. The sequence of purchase and exchange thus ends either when she decides to keep a product or when the product is returned without any future purchase.

3.2. Consumer's Optimization Problem

At the beginning of the search, the consumer can choose whether to buy the first product (and decide which product to choose) or walk away. Afterward, each visit to the retailer is associated with one product that the consumer has already bought and experienced (so that ξ_j has been observed): at that stage, she can choose to (i) keep the product, (ii) return the product and leave without any purchase, or (iii) exchange it for her next choice (also to be decided). Because there is a finite sequence of products that are only bought once, the consumer problem can thus be written as a finite-horizon stopping problem with horizon length $|J|$:

$$\bar{V}(A) = \max_{i \in A} \{E_{U_i}[V(i, A, U_i)]\}, \quad (2)$$

$$V(i, A, U_i) = \max \left\{ \underbrace{U_i}_{\text{Keep product}}, \underbrace{-f_i + \max \left\{ \overbrace{0}^{\text{Leave}}, \overbrace{\bar{V}(A \setminus \{i\})}^{\text{Exchange}} \right\}}_{\text{Return product}} \right\}. \quad (3)$$

In addition, the terminal condition is $\bar{V}(\emptyset) = 0$. The first maximization problem denoted by $\bar{V}(A)$ represents the consumer's maximum expected utility given that a set of products A is available. The second stage $V(i, A, U_i)$ denotes her expected utility once product i has been chosen and utility U_i realized, given in Equation (1). The first term in $V(i, A, U_i)$ represents the

decision to keep product i with utility U_i and end the exchange sequence, the second term denotes the decision to return the current product and leave, and the third term denotes the decision to exchange the current product with one that comes next in the preference list. In particular, because returning is costly, the consumer has to incur the return cost f_i in both cases, although she may enjoy the expected future value-to-go if she decides to exchange. Furthermore, the expected utility for the consumer before the search starts is equal to

$$\max \left\{ \underbrace{0}_{\text{Leave}}, \underbrace{\bar{V}(j)}_{\text{Start search}} \right\}. \quad (4)$$

Given a certain assortment and pricing policy, we can, without loss of generality, sort the items such that $\frac{e^{w_1}}{1-e^{-f_1}} \geq \dots \geq \frac{e^{w_{|J|}}}{1-e^{-f_{|J|}}}$. The next lemma describes the solution to the purchase choice problem.

Lemma 1. *Consumers optimally search products in decreasing order of product ranking (i.e., the maximizer of Equation (2) is $\min\{i \in A\}$).*

As a result, the DP in Equation (3) greatly simplifies. Replacing the state A in the value function by the current item j , the DP can be written as

$$V(j, U_j) = \max \left\{ \underbrace{U_j}_{\text{Keep product}}, \underbrace{-f_j + \max \left\{ \underbrace{0}_{\text{Leave}}, \underbrace{E_{U_{j+1}}[V(j+1, U_{j+1})]}_{\text{Exchange}} \right\}}_{\text{Return product}} \right\} \quad (5)$$

for $j \in J$.

In addition, the terminal condition is $V(|J|+1, \cdot) = 0$ (i.e., the utility from no purchase is zero). As we can see, because the order of exploration is given by Lemma 1, we suppressed the dependency of $V(j, U_j)$ on the set of items available for exploration A . Furthermore, the expected utility for the consumer before the search starts is equal to

$$V_0 = \max \left\{ \underbrace{0}_{\text{Leave}}, \underbrace{E_{U_1}[V(1, U_1)]}_{\text{Start search}} \right\}. \quad (6)$$

As we can see, this dynamic program still seems complex, but in reality, there are immediate simplifications that can be obtained. First, in Equation (5), the decision whether to leave or exchange the product is simply a comparison between constants, and as a result, they do not depend on U_j . Second, under our distributional assumption on ξ_j , the value of $-f_j + \max\{0, E_{U_{j+1}}[V(j+1, U_{j+1})]\}$ can be found in closed

form. The next theorem fully describes the solution to the consumer's optimization problem.

Theorem 1. *The consumer's optimal decisions can be described as follows:*

(i) Define

$$j_{\max} := \max\{j : \log(e^{w_j} + e^{-f_j}) \geq 0\}. \quad (7)$$

Then, if the consumer decides to return a product j , she will exchange it if $j \leq j_{\max}$ and leave without purchase if $j > j_{\max}$.

(ii) *Consumer's optimal expected utility can be written as*

$$E_{U_j}[V(j, U_j)] = \log \left(\sum_{i=j}^{j_{\max}} e^{w_i - \sum_{k=j}^{i-1} f_k} + e^{-\sum_{k=j}^{j_{\max}} f_k} \right).$$

(iii) *Let $\Psi = \{1, \dots, j_{\max}\}$ denote the consideration set. The conditional probability of consumers keeping product $j \in \Psi$, given that product j has just been purchased and prior to the realization of ξ_j , is*

$$\frac{e^{w_j}}{e^{w_j} + \sum_{i=j+1}^{j_{\max}} \left(e^{w_i - \sum_{k=j}^{i-1} f_k} \right) + e^{-\sum_{k=j}^{j_{\max}} f_k}}.$$

(iv) *The unconditional probability of consumers keeping product $j \in \Psi$ is*

$$P_j = \frac{e^{w_j - \sum_{k=1}^{j-1} f_k}}{e^{-\sum_{k=1}^{j_{\max}} f_k} + \sum_{l \in \Psi} e^{w_l - \sum_{k=1}^{l-1} f_k}}, \quad j \in \Psi, \quad (8)$$

or 0 if $j \notin \Psi$. The probability of leaving the retailer without a product is

$$P_0 = \frac{e^{-\sum_{k=1}^{j_{\max}} f_k}}{e^{-\sum_{k=1}^{j_{\max}} f_k} + \sum_{l \in \Psi} e^{w_l - \sum_{k=1}^{l-1} f_k}}, \quad j \in \Psi. \quad (9)$$

Theorem 1 characterizes the consumer's optimal exploration and the likelihood of her keeping any item. Part (i) of that theorem shows that a consumer's willingness to try a product depends on its net utility and the cumulative return cost she has to bear. Therefore, our model results in a consideration set coming from a sequential search setting. It is similar to the one where consumers form the consideration set up front, as in Wang and Sahin (2018), and then explore all products simultaneously, although it has a different form because there is no recall in our model. In our formulation, we can easily see how the size of consideration set $j_{\max}$ varies with the problem parameters. When returns are free for the consumer (i.e., $f_j = 0$ for all j), then all products are considered $j_{\max} = |J|$. As the return cost increases, the consideration set shrinks, and thus the consumer prefers trying fewer products. In the limiting case when $f_j = \infty$ for all j , the consumer's consideration set shrinks

to only those products that have $w_j \geq 0$. But from part (iv), we can also see that it is the first product considered and tried (i.e., product 1 when $w_1 \geq 0$) that will be kept,⁷ meaning that the other products in the consideration set are disregarded; if there is no product in the consideration set (i.e., $w_1 < 0$), the consumer will not start her search but prefers to leave the retailer without any purchase attempt.

Using part (i) of Theorem 1, the consumer's optimal choice problem simplifies to a trade-off between keeping and exchanging a product for each product in the consideration set. This simplification allows us to write $E_{U_j}[V(j, U_j)]$ in closed form; see part (ii). In particular, the profit-to-go from the products still to be explored is related to the sum of these products' attractiveness, measured as $e^{w_i - \sum_{k=j}^{i-1} f_k}$ if we are currently holding product j . This expression is decreasing in i , meaning that a product that is next on the list will provide higher value compared with one that will be explored later, because the impact of return costs will be higher for the latter. In addition, the drop in attractiveness will be more pronounced as f_i increases.

Given this characterization, we derive in parts (iii) and (iv) the probabilities that a consumer keeps product j . Note from (iii) that in our model the conditional probability of keeping product j may be increasing or decreasing in j , which allows us to capture possible behaviors as those of Samorani et al. (2018). From (iv), the unconditional probabilities have an attraction form, which includes the well-known MNL demand function. In particular, if the consumer's return cost is $f_j = 0$ for all j , the probability of keeping product j looks identical to the MNL demand where product attractiveness is e^{w_j} . Yet the derivation of our choice probabilities is fundamentally different: we obtain them from a *sequential* trial of random utilities with an average that decreases in the order. Interestingly, when product shocks ξ_j are logistic distributed, then we obtain an equivalent formulation to that of a *simultaneous* trial of utilities with Gumbel-distributed shocks, as in the standard MNL.⁸ By contrast, when f_j are high, then the attractiveness of all products except the first one decreases, including that of the outside option. As a result, consumers stick to their first choice more often. In the limiting case $f_j = \infty$, there is no choice per se, and all consumers keep the first product. In summary, our sequential search framework allows us to internalize the search costs in the choice probabilities, which cannot be done easily when product choice is determined simultaneously.

One of the salient points of this choice model is that product order is important, because the term associated with return costs of j , $\sum_{k=1}^{j-1} f_k$, depends on it.

Consider, for instance, two products j_1 and j_2 , with w_{j_1} slightly higher than w_{j_2} . In this case, j_1 will be considered earlier, and the probability of keeping j_1 may be significantly higher than that of keeping j_2 . These order effects have been discussed before in other choice models (Alptekinoğlu and Semple 2016). Furthermore, the shortcomings of the independence of irrelevant alternatives property do not apply to our model, in contrast to the MNL. Indeed, when a product is added between j_1 and j_2 , the ratio of demand for these two products varies. Finally, it is worth highlighting that our structure can support some of the empirical findings of Samorani et al. (2018): higher prices for the first item (i.e., reduced net utility) and larger store assortment both tend to increase return probability; on the other hand, in our model, the conditional probability of exchanging the item is independent of the price of the first item, whereas they find it is increasing in it.

4. Retailer's Optimal Decisions

We can now turn to the retailer's problem. By pricing adequately, the retailer can enforce a particular sequence of search through the items (possibly excluding some), which we denote by $S = (s_1, \dots, s_k)$; that is, product s_1 is tried first, then s_2 , and so on, possibly before s_k (hence it may be the case that the consumer's consideration set does not cover S). Defining $j_{\max}$ from (7) as a function of r , the retailer's expected profit function can thus be expressed as

$$\max_S \max_r \Pi := \sum_{j=1}^{j_{\max}} \left(r_{s_j} - \sum_{i=1}^{j-1} c_{s_i} \right) P_{s_j} - \sum_{i=1}^{j_{\max}} c_{s_i} P_0. \quad (10)$$

Note that for this formulation to be consistent, the search sequence must be optimal for the consumer (i.e., $\frac{e^{w_{s_1}}}{1-e^{-f_{s_1}}} \geq \dots \geq \frac{e^{w_{s_k}}}{1-e^{-f_{s_k}}}$) (Lemma 1). Equation (10) is composed of the profit that the retailer makes when a consumer decides to keep a product minus the total return cost incurred by the retailer up to that choice. From this, we must subtract the cost the retailer has to bear when the consumer decides to leave without making a purchase at the end of her search.

We can now proceed with the analysis of the retailer's optimal assortment, pricing, and return policy. We analyze two scenarios: (i) where prices and return costs are exogenously given and the retailer solely determines his optimal assortment and (ii) where the retailer endogenously determines prices. Equation (10) reveals that although the consumer choice model with varying (and potentially price-dependent) return costs f_j can be formulated, the profit function's structure becomes intractable.⁹ In the rest of the paper, we apply Assumption 1 (i.e., return costs are product independent). Under this assumption, to

have consumers search in sequence S imposes a decreasing order of net utility (i.e., $w_{s_1} \geq \dots \geq w_{s_k}$).

Assumption 1. Return costs for consumers and retailer are constant for all products: $f_j = f$ and $c_j = c$ for $j \in N$. Moreover, let $d = c + f$ be the total return cost, and let $\alpha = f/(c + f)$ denote the consumer share.

4.1. Optimal Assortment Under Exogenous Prices

Assume here that retail prices $r = (r_1, \dots, r_n)$, net utilities $w = (w_1, \dots, w_n)$, total return cost d , and consumer share α are exogenously given. The retailer's goal is to choose the assortment that maximizes his profit. In this case, given an assortment, the sequence of search is simply in order of decreasing net utility; see Lemma 1. Moreover, it is apparent from Theorem 1 that there is no point in offering products $j > j_{\max}$ (even if such products are offered, the consumer will refuse to try them, so they can be removed without loss of generality).¹⁰ We let $x_j = \{0, 1\}$ indicate whether product j is included in the assortment. The assortment optimization problem with fixed return costs reduces to

$$\Pi^{exo} = \max_{x \in \{0,1\}^{j_{\max}}} \left\{ \sum_{j=1}^{j_{\max}} \left[r_j - (1-\alpha)d \left(\sum_{i=1}^{j-1} x_i \right) \right] P_j(x) - (1-\alpha)d \left(\sum_{j=1}^{j_{\max}} x_j \right) P_0(x) \right\}, \quad (11)$$

$$\text{where } P_j(x) = \frac{x_j e^{w_j - (\sum_{i=1}^{j-1} x_i)}}{\sum_{l=1}^{j_{\max}} x_l e^{w_l - \alpha d (\sum_{k=1}^{l-1} x_k)} + e^{-\alpha d (\sum_{j=1}^{j_{\max}} x_j)}}.$$

When the total return cost d is zero, the retailer's assortment problem displayed in Equation (11) is equivalent to one where consumer demand is derived from the MNL choice model. Talluri and Van Ryzin (2004) show that, for this special case, the optimal assortment consists of those products that yield the highest revenue. In this case, the assortment problem can be found in linear time. In the presence of return costs, however, such an appealingly simple structure does not persist, suggesting that it may be

difficult to obtain an optimal assortment. We illustrate this observation with one example.

Example 1. Let $n = 3$, $(w_1, w_2, w_3) = (2.2, 2, 0.1)$, and $r = (r_1, r_2, r_3) = (3, 2, 1.5)$. Furthermore, let the return share be $\alpha = \{0, 1\}$ and the total return costs $d = \{0, 1\}$. Table 1 shows the probability of keeping a product P_j and the retailer's expected profit Π for each assortment J , for both return shares. The optimal assortment is in bold. We can see from the left panel that when $\alpha = 0$, a revenue-ordered assortment is optimal. In this case, because the retail margin is ordered in the same way as product attractiveness, such an assortment is also a popular assortment (one made of the most attractive products for the consumer). In the right panel, where the consumer pays the entire return cost, we see that the retailer's optimal assortment is given by $\{1, 3\}$, which is neither a revenue-ordered set nor even a set made of a connected sequence of items. We can provide some intuition for this result. In this instance, the parameters are chosen so that all products are always considered (i.e., $j_{\max} = |J|$). In the presence of costly returns for the consumer, the retailer tends to add more products to the assortment, thereby reducing the likelihood of the outside option. For example, adding a second product to assortment $\{1\}$ decreases the attractiveness of the outside option from e^{-d} to e^{-2d} . It thus depends only on $|J|$, not on whether the added product is 2 or 3. On the other hand, including more items cannibalizes sales of the existing assortment, and this cannibalization effect is stronger when the newly added products are more attractive. For example, adding product 2 to assortment $\{1\}$ cannibalizes the sales of product 1 more than adding 3. Hence, adding "sufficiently" unattractive products such as product 3 may be beneficial because cannibalization is weak and the effect on the outside option is the same. This turns out to be optimal in this case.

It is clear from this example that in the presence of product return cost, the optimal assortment is not revenue ordered anymore, potentially requiring full enumeration to find the optimal solution. Nevertheless, we are able to develop a simple algorithm to solve the assortment problem using a fractional programming transformation as in Megiddo (1979)

Table 1. Consumer Choice Probabilities P_j and Retailer Expected Profits Π for All the Possible Assortments

J	$\alpha = 0$						$\alpha = 1$					
	P_1	P_2	P_3	P_0	Π	Π	P_1	P_2	P_3	P_0	Π	Π
$\{1\}$	0.90			0.10	2.70	2.60	0.96			0.04	2.88	
$\{2\}$		0.88		0.12	1.76	1.64		0.95		0.05	1.91	
$\{3\}$			0.52	0.48	0.79	0.31			0.75	0.25	1.13	
$\{1, 2\}$	0.52	0.42		0.06	2.40	1.86	0.76	0.23		0.01	2.74	
$\{1, 3\}$	0.81		0.10	0.09	2.58	2.30	0.94		0.04	0.01	2.89	
$\{2, 3\}$		0.78	0.12	0.11	1.73	1.40		0.93	0.05	0.02	1.94	
$\{1, 2, 3\}$	0.49	0.40	0.06	0.05	2.35	1.67	0.76	0.23	0.01	0.00	2.74	

or Rusmevichientong et al. (2010). The original problem given in Equation (11) can be written as $\max \theta$ s.t. $\Pi^{exo} \geq \theta$. Further algebraic manipulation of the constraint yields

$$\max_{x \in \{0,1\}^{j_{\max}}} \left\{ \sum_{j=1}^{j_{\max}} \left[r_j - (1-\alpha)d \left(\sum_{i=1}^{j-1} x_i \right) - \theta \right] x_j \cdot e^{w_j - \alpha d \left(\sum_{k=1}^{j-1} x_k \right)} - \left[(1-\alpha)d \left(\sum_{j=1}^{j_{\max}} x_j \right) + \theta \right] \cdot e^{-\alpha d \left(\sum_{j=1}^{j_{\max}} x_j \right)} \right\} \geq 0. \quad (12)$$

Equation (12) is composed of a sum of (nonlinear) terms. Thus, for any given real value θ , one can assess the feasibility of the inequality via dynamic programming, as shown in the following lemma.

Lemma 2. For any $\theta \geq 0$, $K \leq j_{\max}$, item $j \in \{1, \dots, j_{\max}\}$, and position $p \in \{1, \dots, K\}$, let $v_{\theta,K}(j, p) = (r_j - (1-\alpha)d(p-1) - \theta)e^{w_j + \alpha d(K-p+1)}$. Consider the following dynamic program (one for each K) where $H_{\theta,K}(j, K+1) = 0$ for any $j \leq j_{\max}$:

$$H_{\theta,K}(j, p) = \begin{cases} \max_{j' < j \leq j_{\max}} \{v_{\theta,K}(j', p) + H_{\theta,K}(j', p+1)\} & \text{when } j < j_{\max}; \\ -\infty & \text{otherwise.} \end{cases} \quad (13)$$

Then $\Pi^{exo} \geq \theta$ if and only if $H(\theta) := \max_{K \in \{1, \dots, j_{\max}\}} \times \{H_{\theta,K}(0, 1) - (1-\alpha)dK\} \geq \theta$. Moreover, $H(\theta)$ is decreasing in θ .

The lemma thus provides an efficient way to find the optimal assortment for any θ . To ensure that this procedure is efficient, the DP in (13) must be easy to solve: for any θ and K , the suggested DP turns out to be computationally efficient, involving $O(n^2)$ nodes and $O(n)$ searches in each node; because this needs to be tried for all $K \in \{1, \dots, j_{\max}\}$, $H(\theta) \geq \theta$ can be checked in $O(n^4)$ for each θ . We can use this observation in Algorithm 1, which evaluates a small number of candidate values for θ , in powers of $(1-\epsilon)$. The algorithm thus develops an approximation to the original problem with a revenue at least $1-\epsilon$ times the optimal revenue, as shown next.

Theorem 2. Algorithm 1 returns an assortment that yields a profit of $\theta^* \geq (1-\epsilon)\Pi^{exo}$. Its computational complexity is $O(n^4 m)$, where m is defined in Algorithm 1.

Theorem 2 provides a multiplicative approximation scheme that is polynomial in n and m . Note that m depends on the input data, through the probability of the outside option $P_0(J_0)$, and hence it is not a fully polynomial-time algorithm scheme. Fortunately, we

Algorithm 1

INPUT: $r, d, \alpha, w, \epsilon$.

- 1: Compute $H_0 := H(0)$.
- 2: Let J_0 be the corresponding optimal assortment.
- 3: Let $m = \left\lceil \frac{\log(P_0(J_0))}{\log(1-\epsilon)} \right\rceil$ where $P_0(J_0)$ is defined in Equation (9).
- 4: **if** $H_0 \leq 0$ **then**
- 5: No assortment that provides positive profit. Return solution $J^* = \emptyset$ and $\theta^* = 0$.
- 6: **else**
- 7: **for** k from 1 to m , **do**
- 8: $\theta_k = H(0)(1-\epsilon)^k$.
- 9: Solve $H_{\theta,K}(0, 1)$ for each K as defined in (13) and compute $H_k := H(\theta_k)$.
- 10: Let J_k be the corresponding optimal assortment.
- 11: **end for**
- 12: Let $k^* = \max\{k | H_k \geq \theta_k\}$. Return solution J_{k^*} and $\theta^* = \theta_{k^*}$.
- 13: **end if**

have that $-\log(P_0(J_0)) \leq \log(1 + ne^{w_1 + n(1-\alpha)d})$, which suggests that the complexity is very reasonable provided that w_1 is not too large. In our numerical experiments, Algorithm 1 reached solutions extremely quickly.

4.2. Optimal Assortment and Pricing

We now consider the case when the retailer can jointly optimize over his assortment as given in Equation (10). In this setting, for a given assortment, there may be several search sequences to consider depending on how prices are set. It turns out that the sequence that maximizes Π is the one in which utilities are decreasing, as shown next. This implies that the consumer choice formulation derived in Section 3 does not need to be modified as prices change.

Lemma 3. When the retailer can optimize over prices, it is optimal to price in such a way that consumers search products in decreasing order of utility u_j . In other words, if the offered assortment is $J = \{s_1, \dots, s_k\}$ such that $u_{s_1} \geq \dots \geq u_{s_k}$, then $S^* = (s_1, \dots, s_k)$.

As a result of the lemma, we can without loss of generality restrict our attention to sequences $S = (s_1, \dots, s_k)$, where $u_{s_1} \geq \dots \geq u_{s_k}$. The retailer's maximization problem can hence be expressed as $\Pi^* = \max_S \Phi(S)$, where

$$\Phi(S) := \max_r \Pi = \sum_{j=1}^{j_{\max}} \left(r_{s_j} - (1-\alpha)d(j-1) \right) P_{s_j} - (1-\alpha)d j_{\max} P_0 \quad (14)$$

$$\text{s.t. } e^{u_{s_1} - r_{s_1}} \geq e^{u_{s_2} - r_{s_2}} \geq \dots \geq e^{u_{s_k} - r_{s_k}}, \quad (15)$$

where the choice probabilities P_{s_j} are defined in Equation (8) and $j_{\max}$ is a function of $\mathbf{r}$, defined in (7). The next theorem describes the retailer's optimal assortment and pricing decisions in this case. It requires the use of $W(\cdot)$, the Lambert W function.

Theorem 3. *For any sequence S such that $u_{s_1} \geq \dots \geq u_{s_k}$, the optimal pricing policy has the following structure. There exists $0 \leq j_{\min} \leq j_{\max}$ so that*

- *For $j \leq j_{\min}$, the retailer “prices for margin”: $r_{s_j}^* - (1 - \alpha)d(j - 1) = r_{s_1}^* \geq 0$ provided that $r_{s_j}^* < u_{s_j} - \log(1 - e^{-\alpha d})$.*
- *For $j_{\min} < j \leq j_{\max}$, he “prices for search”: $r_{s_j}^* = u_{s_j} - \log(1 - e^{-\alpha d}) \geq 0$.*
- *For $j > j_{\max}$, he “prices for exclusion”: $r_{s_j}^* > u_{s_j} - \log(1 - e^{-\alpha d})$ so that $P_{s_j} = 0$.*

Moreover, when $j_{\min} \geq 1$, $r_{s_1}^* = 1 + \pi(j_{\min}, j_{\max})$, where $\pi(j_{\min}, j_{\max}) = \Phi(S)$ is defined in (14) and can be rewritten as

$$\pi(j_{\min}, j_{\max}) = W \left[e^{\beta(j_{\min}, j_{\max})} \left(\sum_{j=1}^{j_{\min}} e^{u_{s_j} + d(j_{\min} + 1 - j) - 1} \right) \right] - (1 - \alpha)dj_{\min} - \beta(j_{\min}, j_{\max}), \quad (16)$$

with

$$\beta(j_{\min}, j_{\max}) = - \sum_{j=j_{\min}+1}^{j_{\max}} \left(u_{s_j} - \log(1 - e^{-\alpha d}) - \frac{(1 - \alpha)d}{e^{\alpha d} - 1} \right) \cdot (1 - e^{-\alpha d})e^{-(j-1-j_{\min})} < 0. \quad (17)$$

Furthermore, $S^* = (1, \dots, n)$ is an optimal sequence, and the optimal effective assortment $\{1, \dots, j_{\max}\}$ is a popular set.

Theorem 3 thus characterizes the optimal pricing and assortment decisions. Specifically, it shows that for any chosen assortment, and hence a sequence S of cardinality k , the pricing structure is driven by two indices, $j_{\min}$ and $j_{\max}$, and that the corresponding profit $\pi(j_{\min}, j_{\max})$ can be computed as a function of those, from Equation (16). Such a function increases in product utilities, and hence it is optimal to offer the k most attractive products (with the highest u_j) (i.e., a popular set). In other words, it is sufficient to consider sequences of the form $S = (1, \dots, k)$ where items have been sorted in order of decreasing utility u_i . In particular, because products can always be dropped by pricing them high enough, there may be multiple optimal sequences when $j_{\max} \leq k$, although the effective set of products tried by consumers will be equal to $\{1, \dots, j_{\max}\}$ for all of them.

When the return cost for the consumer is free, $\alpha = 0$, then $\beta(j_{\min}, j_{\max}) = 0$, and the consumer's consideration set constraint is never binding (i.e., $j_{\min} = j_{\max} = k$). In this case, even though the retailer advertises free returns, he in fact folds the return cost into the price. We call this a “price for margin” scheme, where return costs are effectively hidden in the price so that the

retailer margin for each item remains identical despite these return costs. This result resembles the one of multiproduct pricing under the multinomial logit model, where the same markup is found to be optimal (Anderson et al. 1992). In the context of product returns, such a pricing scheme also means that the consumer effectively pays only the return cost when she ends up keeping one of the products. On the other hand, if she leaves without a final purchase, the entire cost is borne by the retailer. The resulting optimal profit can thus be expressed in this case through the Lambert W function, as in Li and Huh (2011) or Heese and Martínez-de Albéniz (2018), but in contrast with these, the assortment's net attractiveness $\sum_{j=1}^k e^{u_{s_j} + d(k+1-j) - 1}$ internalizes the potential return cost. Furthermore, an additional cost to the retailer also appears as a factor $-dk$ that is added to the Lambert W term.

On the other hand, when $\alpha > 0$, we observe that $j_{\max} \leq k$. Indeed, given that return costs are costly for the consumer, she may prefer not to try all the products at the retailer's optimal price. Moreover, there exists a second index $j_{\min}$ that separates the products. For those below $j_{\min}$, the “price for margin” structure is optimal, as when $\alpha = 0$. For those above it, the consideration set constraint $w_{s_j} \geq 1 - e^{-\alpha d}$ becomes binding so that $r_{s_j}^* = u_{s_j} - \log(1 - e^{-\alpha d})$; we call this pricing structure “price for search” so that prices are set low enough to ensure that the consumer tries the product, even if she would be indifferent between this and returning the product without any further exchange.¹¹ Finally, there will be products for which price for search leads to negative margins. In that case, it will be optimal to take these products out of the assortment. Because the retailer has chosen to have them in the choice set, they can still be effectively removed by setting the price high enough, in the “price for exclusion” regime.

This pricing structure has an interesting feature: on the one hand, prices are increasing with $j \leq j_{\min}$ because $r_{s_j}^* = r_{s_{j-1}}^* + (1 - \alpha)d$, as a result of the additional return costs incurred by later items; on the other hand, prices are decreasing with $j \in [j_{\min} + 1, j_{\max}]$, because later products offer less utility to the consumer, and thus the retailer cannot charge as much. Therefore, the optimal structure is unimodal, with a peak at $j_{\min}$ or $j_{\min} + 1$. Interestingly, Wang and Sahin (2018) find a similar pricing scheme when consumers form a consideration set. In their case, prices are set equally, except for the price of a final, single product. This ensures that consumers include this item in the consideration set. We use a very different sequential setting, but our structure is similar in spirit: we find that prices are optimally set at equal margins for some of the items (those $j \leq j_{\min}$) and that some products (possibly none or more than one) will be potentially

priced at the right level so that they are at least tried, even though consumers are indifferent between trying them and leaving. This feature was also revealed in a setting where consumers sequentially search prior to sales (Najafi et al. 2016).

As a consequence of Theorem 3, to obtain the optimal assortment, we need only to maximize $\pi(j_{\min}, j_{\max})$ over $j_{\min}$ and $j_{\max}$. It turns out that the optimal values for $j_{\min}$ and $j_{\max}$ can be described in closed form, as shown next.

Theorem 4. The optimal profit $\Pi^* = \max_{j_{\min}, j_{\max}} \pi(j_{\min}, j_{\max})$ can be characterized by

$$j_{\min} = \max\{j | u_j \geq 1 + \pi(j, \bar{j}) + (1 - \alpha)d(j - 1) + \log(1 - e^{-\alpha d})\}, \quad (18)$$

$$j_{\max} = \max\{j_{\min}, \bar{j}\}, \quad (19)$$

where $\pi(\cdot)$ was defined in Theorem 3 and

$$\bar{j} = \max\left\{j | u_j \geq \log(1 - e^{-\alpha d}) + \frac{(1 - \alpha)d}{e^{\alpha d} - 1}\right\}. \quad (20)$$

Moreover, there are two possible regimes that the retailer may want to follow. There exists $\bar{\alpha}$ such that

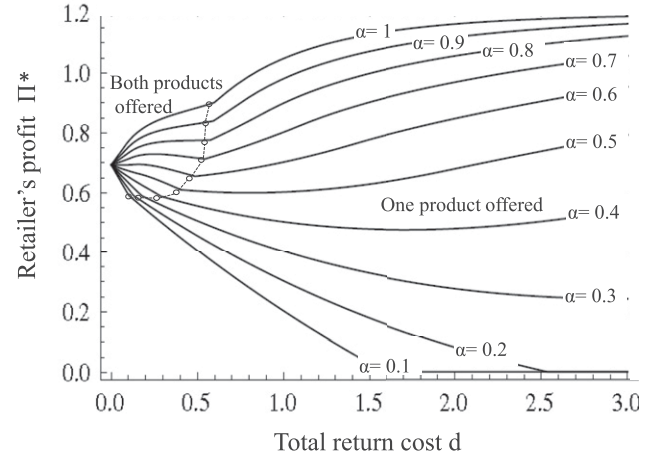
- Case 1. When $\alpha \leq \bar{\alpha}$, $j_{\min} = j_{\max}$, and $j_{\max}$ increases with α .
- Case 2. Otherwise, $j_{\min} < j_{\max}$, and $j_{\max}$ increases in α , whereas $j_{\min}$ decreases in α .

The theorem thus completely describes the structure of the optimal solution and how it varies with α . In contrast with the problem with exogenous prices, allowing the retailer to set prices optimally makes the problem with repeated returns computationally more tractable. Specifically, finding the optimal assortment only requires inspecting the n candidates $(j_{\min}, j_{\max}) \in \{(0, \bar{j}), (1, \bar{j}), \dots, (\bar{j}, \bar{j}), (\bar{j} + 1, \bar{j} + 1), \dots, (n, n)\}$. This leads to a complexity of $O(n)$ because the required computations of $\beta(j_{\min}, j_{\max})$ and $\pi(j_{\min}, j_{\max})$ in Equations (16) and (17) can be performed recursively.

We can now study how retailer optimal decisions and profit change with the return policy parameters. Figure 2 depicts the retailer's optimal profit as a function of the total return cost d for varying α , with two products and utilities, $u_1 = 1.2$ and $u_2 = -0.8$.¹²

We see that the optimal assortment shrinks when d increases. We can also assess the impact of the return share α : increasing it leads to a larger assortment. This result is intuitive: the less the retailer pays for returns, the more he is willing to offer because the net cost to him is decreased. This is actually a direct implication from Theorem 4. The notion that consumers sharing more of the burden of returns allows the retailer to take more risks with assortment selection, or to offer product variety more liberally, was also found in setting where consumers can only exchange products once (Alptekinoglu and Grasas 2014). Moreover,

Figure 2. Retailer's Optimal Profit as a Function of d for $\alpha = \{0.1, \dots, 1\}$



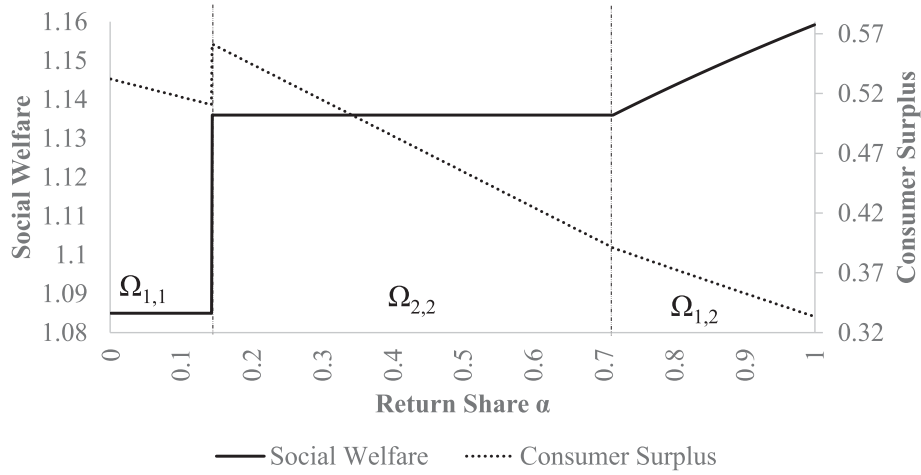
because $j_{\max}$ is determined by the items such that u_j are higher than a continuous function of α from Equations (19) and (20), we obtain the following corollary.

Corollary 1. When $u_1 > u_2 > \dots > u_n$, as α increases infinitesimally, the assortment breadth $j_{\max}$ remains the same or increases by 1.

The impact on optimal profits is less intuitive: we find that an increase in the total return cost d does not necessarily decrease profits and prices.¹³ This can be explained by observing that a higher return cost will increase a consumer's probability of keeping more preferred products, and thus the likelihood of leaving the retailer without a purchase decreases. The role of α , on the other hand, is unambiguous: profits increase in α . We analytically explore next the impact of α on retailer profits, consumer surplus, and social welfare.

4.3. Impact of Return Share on Social Welfare and Consumer Surplus

Figure 2 suggests that the retailer's profit improves as a larger share of the return cost is passed on to the consumer. But if this is the case, it is not clear whether such a decision is beneficial for consumers or social welfare. Recalling that V_0 is the expectation of consumer surplus (see Equation (4)), we can define social welfare Z as the sum of the retailer's expected profit and the consumer's expected surplus: $Z = \Pi^* + V_0$. We proceed with an illustration of social welfare and consumer surplus in Figure 3, for a two-product example with $u_1 = 1.2$, $u_2 = -0.5$, and $d = 0.15$. The region $\Omega_{j_{\min}, j_{\max}}$ denotes the optimal choices $(j_{\min}, j_{\max})$ for the retailer. Figure 3 shows that for each α in region $\Omega_{j_{\min}, j_{\max}}$, the retailer chooses $(j_{\min}, j_{\max})$ as the optimal assortment and pricing strategy following Theorem 4. We can see that within each region, the consumer (dotted line) will be worse off as the return share α

Figure 3. Social Welfare (Solid Line) and Consumer Surplus (Dotted Line) as a Function of α 

increases. However, a consumer also prefers more variety over less, which the retailer is only willing to offer if the consumer shares part of the return cost. Therefore, as we move across regions, consumer surplus may jump up: we see an upward discontinuity between regions $\Omega_{1,1}$ (where one product is offered and priced for margin) and $\Omega_{2,2}$ (where two products are offered and both are priced for margin). Note that such a jump may or may not exceed the loss since the previous jump. As a result, consumer surplus is, in general, nonmonotonic in α , although it is decreasing when continuous.

Regarding social welfare (solid line), we see that as long as the retailer can price all products for margin (i.e., regions $\Omega_{1,1}$ and $\Omega_{2,2}$), social welfare is flat: from a welfare perspective, it does not matter who pays the return cost. However, because the consumer experiences a jump when more products are offered, social welfare increases when we transition from $\Omega_{1,1}$ to $\Omega_{2,2}$. Moreover, when the retailer prices for search, he gains more than the consumer loses, resulting in a total increase in welfare. In summary, we find that although a consumer will lose when the return share is high, it is still optimal for society at large to let a consumer pay the return share. We formalize this observation in the following theorem.

Theorem 5. *As the return share α increases, we have the following:*

- Case 1. When $\alpha \leq \bar{\alpha}$ (defined in Theorem 4), retailer profit Π^* increases; consumer surplus V_0 decreases except at points where $j_{\min} = j_{\max}$ change, in which case V_0 jumps up; and social welfare Z is constant except at points where $j_{\min} = j_{\max}$ change, in which case Z jumps up.
- Case 2. Otherwise, retailer profit Π^* increases, consumer surplus V_0 decreases, and social welfare Z increases.

Theorem 5 is based on three effects that occur when the retailer increases the return share. First, when all

prices can be set optimally and the retailer does not alter the assortment, society is indifferent about the return cost split between the consumer and the retailer. However, when α increases, the retailer may offer a larger assortment. This greater variety is good for consumer surplus and also increases social welfare. Finally, when the return share increases further, the retailer needs to revise his prices downward, which encourages the consumer to keep products earlier (because her net utility increases). This both reduces overall returns and improves social welfare. For these reasons, passing all costs related to returns to the consumer increases welfare and could even benefit the consumer (if this expands the assortment). This is in line with Su (2009), who shows that the retailer is better off when giving partial refunds (compared with full refunds), and in opposition with Che (1996) or Davis et al. (1995), who find that free returns are best for society, particularly in models where assortment is fixed and there is a salvage value higher than the return cost. By contrast, in our model, a return creates a cost that hurts social welfare independent of who pays the cost. However, when consumers internalize such costs, prices are reduced and variety increases, and this is why it is beneficial for welfare.

5. Conclusion and Extensions

In this paper, we have developed a framework where return and exchange patterns can be incorporated into a sequential consumer choice model. It is based on the premise that consumers may not be able to resolve uncertainty prior to purchase and will thus sequentially try products, perhaps returning or exchanging them if they are unsatisfied. The consumer problem can be solved using dynamic programming, and it generates choice probabilities that depend on product attractiveness, the rank in which

the consumer would try that item, and the return costs that would have to be paid if a return or exchange takes place. When uncertainties follow a logistic distribution, these probabilities can be written in a closed-form attraction demand model.

We embed this demand model into a decision problem where a retailer can optimize prices, assortment, and the share of return costs to be paid for by the consumer. We first show that when product prices are given, the assortment problem is challenging, but we develop an approximation scheme that provides a solution that is at least a multiple of the optimal solution. When prices can be adjusted, we are able to characterize the structure of the optimal solution: the optimal assortment is a popular set, in which the more popular products will be priced for margin, so that their revenue net of return costs will be the same, whereas the less popular products will be priced for search, so that their revenue net of costs will be low enough to be added to the consumers' consideration set. This dual structure will allow the retailer to maximize profits while reducing return costs, and optimal solutions can be computed efficiently. Finally, we show that passing the return cost onto the consumer is beneficial for the retailer and could also improve consumer surplus if it incentivizes the retailer to offer more variety. In addition to the base case model, we also advance the optimal pricing problem when consumers are heterogeneous in their preference and search costs for a two-product setting.

Our model can be extended in several directions. First, we have considered the time to evaluate a product to be sufficiently long to resolve all uncertainty. In many cases, however, return policies may not only differ in terms of costs but also in terms of return length. It would be interesting to capture these aspects in a consumer choice model. Second, we have considered a monopolistic retailer, whereas in some practical contexts, retail competition on return policies is key. Developing a model that captures competing retailers would help improve our understanding of how assortments, pricing, and return fees should be set in the presence of strategic interactions. Third, we have assumed that consumers can only purchase and return one product at a time. For some product categories, though, such as low-cost and low-volume items (i.e., T-shirts or shoes), consumers may purchase more than one item with the intent of keeping only a few. Incorporating this feature requires considering simultaneous learning of the random product utilities and requires a different framework, possibly simplifying distributional assumptions on the random shocks. Finally, our tractable analytical model is amenable to empirical testing as in Samorani et al. (2018), and such validation would be of great value in shedding light on the sensitivity of consumers to return costs.

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Endnotes

¹ Product independent return costs include hassle costs, transportation costs, or a total restocking fee independent of the price of the product.

² Numerical experiments carried out with other types of distributions suggest that the qualitative features of the demand model remain similar. However, the implications for assortment and pricing may change (e.g., price differences could be higher or lower compared with our results).

³ The no-recall assumption is based on the premise that customers do not repeatedly buy and return the same product multiple times. Meanwhile, the search cost is associated with the cost of returning the product and its consequently being out of reach. In a numerical study, we relaxed the no-recall assumption and found no significant difference with our base model when the return costs are positive (even when they are small).

⁴ Here, r is taken as an input for the customers, although it may be a decision taken later by the retailer.

⁵ As noted in Endnote 2, other types of distribution yield similar qualitative features of the demand model, but the assortment and pricing implications might differ.

⁶ Although it is theoretically possible for consumers to purchase a product previously returned (recall), incorporating such a feature is likely analytically intractable. Also, as noted in Endnote 3, we did not find significant differences between our base model and one that includes recall for cases where return costs are positive (even when they are small).

⁷ When $w_1 \geq 0$, in Equation (8), the denominator tends to e^{w_1} when f_j tends to ∞ , and hence $P_1 \rightarrow 1$.

⁸ See Adams and Messick (1958), Yellott (1977), and figure 2.4 in Anderson et al. (1992) for different derivations of the MNL choice probabilities.

⁹ For instance, product attractiveness of product s_i is equal to $w_{s_i} - \sum_{j=1}^{i-1} f_{s_j}$ and hence depends on which products were included in the assortment that appear before s_i in the search sequence. Moreover, when f_{s_j} is proportional to r_{s_j} (e.g., proportional stocking fees), then the price of product $s_j < s_i$ affects the attractiveness of s_i . This makes the analysis intractable.

¹⁰ In this case $j_{\max}$ is independent of the search sequence S ; see Equation (7). By contrast, in Section 4.2 where retail prices are a decision variable, $j_{\max}$ varies with retailer decisions with respect to r .

¹¹ Note that one could price ϵ lower to avoid ties, for infinitesimal ϵ . Hence, the tie decision rule does not influence the results.

¹² Note that consumers may still try a product even if u_j is negative. To provide some intuition of why this can be the case, assume that return is free for the consumer ($ad = 0$). In this situation, she would actually prefer to try a product with negative utility (in case all previous products have been returned) in the hope that the random shock is positive, because she could later reverse her decision at no cost.

¹³ It is nevertheless possible to show that when $\alpha = 0$, profits are decreasing in d .

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